

PENUMBRA

Low-Level Design (LLD)

Document ID: PEN-LLD-001 **Version:** 1.0 **Date:** 23 July 2026
Traces to: PEN-HLD-001, PEN-SRS-001

1. Repository Layout

```
penumbra/
├── documents/          SRS/HLD/LLD/QA (src html + pdf), diagrams/, pitch/
├── backend/
│   ├── requirements.txt fastapi, uvicorn, pydantic-settings, numpy, httpx, python-dateutil, p
│   ├── logs/           penumbra.log (rotating, git-ignored)
│   └── app/
│       ├── main.py     app factory, middleware, handlers, startup pipeline, SPA serving
│       ├── core/       config.py · logging_config.py · exceptions.py
│       ├── models/     schemas.py (Pydantic v2 DTOs)
│       ├── logic/      series.py · f107_forecast.py · kp_forecast.py
│       │               · calibration.py · drag.py                [pure, no I/O]
│       ├── service/    swpc_client.py · cache_service.py · forecast_service.py
│       │               · pipeline_service.py · briefing_service.py
│       ├── api/        routes.py
│       └── data/        sample_f107.json · sample_kp.json · sample_noaa45.json · cache.db (ru
├── backend/tests/     test_f107_forecast.py · test_kp_forecast.py · test_calibration.py
│                       · test_drag.py · test_api.py
└── ui/                Vite + React + TS (src/api, src/components, src/pages, src/styles)
```

2. Core (app/core)

2.1 config.py

```
class Settings(BaseSettings):          # env prefix PENUMBRA_, .env supported
    app_name = "PENUMBRA"; version = "1.0.0"
    demo_mode: bool | None = None      # None -> auto: demo unless PENUMBRA_LIVE=1
    anthropic_api_key: str | None = None
    data_dir / log_dir: Path
    obs_ttl_seconds = 10800             # 3 h (CON-4)
    forecast_ttl_seconds = 21600        # 6 h
    max_lead_days = 45
    kp_max_lead_days = 7
    f107_floor_sfu = 64.0               # quiet-Sun minimum (FR-F107-3)
    backtest_min_history_days = 400
    pipeline_refresh_seconds = 10800
```

2.2 exceptions.py

Class	code	HTTP
PenumbraError (base)	PENUMBRA_ERROR	500
ConfigError	CONFIG_ERROR	500
DataSourceError	DATA_SOURCE_ERROR	502
InsufficientDataError	INSUFFICIENT_DATA	422
ForecastError	FORECAST_ERROR	500
NotFoundError	NOT_FOUND	404
ValidationFailure	VALIDATION_ERROR	422

Global FastAPI handlers convert `PenumbraError` → `{"error": {code,message,detail,request_id}}`; a catch-all maps other exceptions to 500 `INTERNAL_ERROR` after logging.

3. Logic Layer (app/logic) — algorithms

3.1 series.py

```
@dataclass DailySeries: dates: list[date], values: np.ndarray # aligned, gap-filled
load_daily_f107(records) -> DailySeries # from SWPC f107_cm_flux json
kp_to_daily(records) -> (DailySeries max_kp, DailySeries ap) # 3-hourly -> daily
# Ap from Kp via the standard ap-from-Kp lookup (Kp thirds -> ap: 0..400)
align(*series) -> common daily grid; forward/interp gap-fill <= 3-day gaps
trailing_mean(values, window) -> np.ndarray # 27-day rotation mean etc.
```

3.2 f107_forecast.py

```
point_forecast(history: DailySeries, lead_days: int) -> np.ndarray
# x0 = last observed; m27 = trailing 27-day mean; mcyc = trailing 90-day mean
# reversion weight w(L) = 1 - exp(-L / TAU), TAU ~ 10 days
# target(L) = blend(m27, mcyc, by L) # short lead -> 27d, long -> cycle
# f(L) = x0 + (target(L) - x0) * w(L) # persistence -> reversion
# floored at settings.f107_floor_sfu

quantile_bands(point: np.ndarray, err_quantiles: dict[int, dict[float,float]])
-> dict[str, np.ndarray] # p05,p25,p50,p75,p95 per lead
# band[q][L] = point[L] + err_quantiles[L][q]; enforce monotone; floor
```

3.3 kp_forecast.py

```
CATEGORIES = ["quiet","unsettled","active","storm"] # Kp <3, 3-4, 4-5, >=5
category_probabilities(daily_kp: DailySeries, lead_days: int, halflife=180)
-> list[dict[str,float]]
# current state s0 = category(last max_kp)
# from history build recency-weighted transition counts state->category@lead
# P(cat | s0, L) = weighted_count / total ; Laplace-smoothed; sums to 1
expected_ap(daily_ap, lead_days) -> (point, lo, hi) # persistence + climatology band
```

3.4 calibration.py — verification core

```
walk_forward_errors(history, lead_days, min_train, step)
    -> dict[int, np.ndarray]                # signed errors per lead (OOS)
    # for origin t in [min_train .. N-lead) step k:
    #     train = history[:t+1]              # STRICT – no future (CON-2)
    #     pred  = point_forecast(train, lead)
    #     err[L].append(history[t+L] - pred[L-1])
    # returns per-lead error arrays -> quantiles feed f107 bands

error_quantiles(errors, qs=(5,25,50,75,95)) -> dict[int, dict[float,float]]
coverage(history, bands_by_origin) -> dict[int, {p50_50, p90}] # empirical coverage
pinball_loss(y, q_pred, q) -> float                        # proper score
skill_vs_reference(pen_rmse, noaa_rmse) -> 1 - pen/noaa      # per lead
reliability(pred_probs, outcomes, bins=10) -> [(pred, obs, n)] # Kp reliability
```

3.5 drag.py

```
density_at(alt_km, f107) -> rho                # reduced model:
    # rho(h,F) = rho0(h) * exp((F - F_ref)/F_scale) # F10.7-scaled exponential atm.
    # rho0(h): tabulated reference density; F_ref=150, F_scale documented
density_band(alt_km, f107_bands) -> {p05,p50,p95}
decay_uncertainty(alt_km, f107_bands, bstar_like, days)
    -> along-track_km {p05,p50,p95}          # da/dt ~ -k*rho*v ; integrate band
    # labelled reduced-model, order-of-magnitude (FR-Drag-3)
```

4. Data Design — SQLite DDL

```
CREATE TABLE IF NOT EXISTS kv_cache (
    key TEXT PRIMARY KEY, payload TEXT NOT NULL,
    fetched_at TEXT NOT NULL, ttl_seconds INTEGER NOT NULL);
CREATE TABLE IF NOT EXISTS forecast_snapshots (
    id TEXT PRIMARY KEY, issued_at TEXT NOT NULL, document TEXT NOT NULL);
CREATE TABLE IF NOT EXISTS pipeline_runs (
    id INTEGER PRIMARY KEY AUTOINCREMENT, agent TEXT NOT NULL,
    started_at TEXT NOT NULL, finished_at TEXT, status TEXT NOT NULL,
    items INTEGER DEFAULT 0, error TEXT);
```

5. API Contract (app/api/routes.py)

Method & Path	Query	200 response (summary)
GET /api/v1/health	—	{status, version, data_mode, uptime_s}
GET /api/v1/forecast/f107	lead≤45	{issued_at, lead_days, point[], bands{p05,p25,p50,p75,p95}[], noaa_point[], history[], data_mode}
GET /api/v1/forecast/kp	lead≤7	{issued_at, days:[{lead, date, probs{quiet,unsettled,active,storm}, expected_ap{p05,p50,p95}}]}
GET /api/v1/forecast/drag	alt_km	{altitudes:[{alt_km, density{p05,p50,p95}, decay_along_track_km{p05,p50,p95}}], note}
GET /api/v1/calibration	—	{coverage_by_lead[], pinball_by_lead[], skill_vs_noaa[], kp_reliability[], summary}
GET /api/v1/observations/f107	days≤3650	{dates[], values[], unit:"sfu"}
GET /api/v1/briefing	—	{briefing, source:"ai" "template"}
GET /api/v1/pipeline/status	—	{data_mode, agents: [{id,name,status,last_run,duration_ms,items,error}]}

422 validation errors are reshaped into the envelope by an override of FastAPI's `RequestValidationError` handler; every response carries `X-Request-Id` .

6. Service Layer Detail

6.1 swpc_client.py

```
BASE = "https://services.swpc.noaa.gov"
fetch_f107()      -> json    /json/f107_cm_flux.json
fetch_kp()        -> json    /products/noaa-planetary-k-index.json
fetch_noaa45()    -> json    /json/45-day-forecast.json
fetch_history()   -> json    /json/solar-cycle/observed-solar-cycle-indices.json
# httpx, timeout 30 s, 2 retries exp backoff; failures -> DataSourceError
```

6.2 forecast_service.py — pipeline (sequence)

```
refresh():
    # startup + scheduled
    1 pipeline.stage("swpc_sync")      -> cache.get_or_fetch(each source, loader|demo)
    2 pipeline.stage("history")        -> series.load + kp_to_daily + align + gapfill
    3 pipeline.stage("backtest")       -> walk_forward_errors + error_quantiles + Kp stats
    4 pipeline.stage("forecast")       -> f107 point+bands, kp categories, expected Ap
    5 pipeline.stage("calibration")    -> coverage, pinball, skill vs NOAA, reliability
    6 pipeline.stage("publish")       -> persist snapshot; expose read-model
    each stage: try/except -> record status/items/error; degrade to last-known-good;
    the pipeline never aborts the app (FR-ING-6).
```

6.3 briefing_service.py

If anthropic_api_key set: Claude Messages API (claude-opus-4-8 , thinking: {type:"adaptive"} , max_tokens 400, 10 s timeout) summarising the current forecast + calibration into an operator note; on any failure log WARN and return a deterministic template. Never raises to caller.

7. UI Design (ui/)

Module	Detail
src/api/client.ts	fetch wrapper: JSON, error-envelope unwrap → typed ApiError ; single retry on 5xx.
components/FanChart.tsx	Canvas: observed history line, point forecast, nested p25–p75 & p05–p95 fills, NOAA overlay dashed; hover readout with tabular-nums.
components/KpStrip.tsx	Per-day stacked category bars (quiet→storm), storm probability emphasised.
components/DragPanel.tsx	Density & along-track decay uncertainty at reference altitudes; reduced-model caveat.
pages/CalibrationPage.tsx	Coverage-by-lead bars vs target line, reliability diagram (Canvas), skill-vs-NOAA curve.
pages/PipelinePage.tsx	Embeds public/diagrams/pipeline-flow.html (iframe) + postMessage live status.
styles/	bg #070a12, panel rgba(18,14,8,.85), solar accent #ffb454 / plasma #ff7a45, magnetosphere teal #4fd4c4; tabular-nums; product blurb + Apache-2.0 in footer (FR-UI-5).

8. Test Plan

Suite	Cases
test_f107_forecast	point reverts toward 27-day mean as lead grows; bands monotone $p_{05} \leq \dots \leq p_{95}$; floor at 64 sfu enforced; synthetic sinusoid-plus-trend recovered within tolerance.
test_kp_forecast	category probabilities sum to 1 per lead; elevated recent activity raises storm probability; Ap band ordered $lo \leq point \leq hi$.
test_calibration	walk-forward is leakage-free (shuffling the future cannot change past-origin errors); coverage of a Gaussian synthetic $\approx$ nominal; pinball loss zero at perfect quantile; skill sign correct.
test_drag	density monotincreasing in F10.7; higher F10.7 band widens decay band; altitude ordering (lower alt $\rightarrow$ denser).
test_api	TestClient: /health; /forecast/f107 contract + band monotonicity; /calibration shape; unknown route 404 envelope; lead>45 $\rightarrow$ 422 envelope; pipeline status shape.